Same Direction, Different Noise

The full record of the project: the problem, the literature, every experiment with its reason, design and result, the methods, and where we stand

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Abstract

This document records the whole project for a reader who has not followed it: how language models are trained with reinforcement learning on verifiable rewards (RLVR), what the literature claims and where it has gaps, and every experiment we ran, each with its question, reason, pre-registered design, result and consequence. It covers an earlier idea (STOP) and why it was dropped; a theory of the mean and second moment of correctness-only updates; the measurements and tests that led to the main diagnosis (advantage rules trace one accuracy–diversity trajectory and differ in progress rate, with noise as a sufficient cause); the answer-key audit; the first method (coverage-weighted replay, about +1.2 points); the 2026 literature and what it changed; base-seeded replay and its controls; the answer-budget check; the matched-accuracy audit of 2026 methods; a pilot of mechanisms that push signal onto rarely solved problems; a transfer probe; a frozen per-problem model of the frontier and its limits; the long Dr. GRPO run, which shows that groups of 32 add speed rather than diversity; rollout allocation by frontier value (on the curve) and external human solutions (not interpretable at matched accuracy; their 512-token advantage reflected the answer budget). Version 5 adds the move to AWS and a vLLM evaluator; the 7B replication at 2,048 tokens and the 1.5B replication at 1,024 tokens; human-solution guidance inside RL (on the curve) and minimum-help credit (not selected); a test of the shared-update-direction explanation with four pilots that change the update's direction or add verified teacher solutions (none qualifies); an exploratory analysis describing similar per-problem profiles across methods at matched pass@1 and attributing, by accounting, about a third of the gain to finishing answers; a restricted comparison with the budgeted-search law; and the two experiments running now (longer and stronger training; published checkpoints and a forced-answer test). Appendices give every method's definition and settings and an experiment tracker.

The idea in one paragraph

RLVR methods that change how sampled answers are scored are usually compared at equal training steps, where a method that progresses more slowly is less sharpened and so has higher pass@k. Compared instead at matched pass@1, the advantage rules we tested lie on one accuracy–diversity trajectory on two model families: each rule's expected update on a problem is a multiple of the gradient of its success probability, and the rules differ mainly in noise, which sets the progress rate. Because pass@k at matched pass@1 measures how evenly success is spread across problems, moving the frontier needs signal on rarely solved problems, which a group of 8 rollouts rarely provides. Interventions that add the policy's or the base model's own samples mostly add speed; whether allocating rollouts by frontier value, or adding information the base model lacks, moves the frontier is what the current experiments decide.

Contents

1 The problem in plain words 4

2 Background you need 4

3 The literature, and where the gaps are 5

4 The earlier idea (STOP) and why it failed 7

5 What is each generated answer worth? (WORTH) 8

6 Measuring the four numbers 9

7 First tests of the theory 10

8 The literature check and the change of direction 12

9 PSR against NSR (Pre-registration 4) 13

10 Matched accuracy, and the NSR paper's own implementation 14

11 Same average, graded noise (Pre-registration 7) 16

12 Changing the objective, and grading the reasoning (Pre-registration 8) 16

13 What the Qwen experiments say together 17

14 An independent review, and what we changed 17

15 A second model family (Pre-registration 9) 18

16 Is the noise the cause? (Pre-registration 10) 19

17 Is the curve a temperature effect? (Pre-registration 11) 20

18 Checking the answer keys 21

19 From diagnosis to method (Pre-registration 12) 22

20 The 2026 literature, and what it changed 23

21 Seeding the memory from the base model (Pre-registration 14) 24

22 Checking the answer budget (Check 16) 25

23 2026 methods at matched accuracy (Pre-registration 15) **26**

24 Pushing signal onto rare problems inside RL (Pre-registration 16, pilot) **26**

25 Does training transfer to rarely solved problems? (Check 17) **27**

26 A model of the frontier (Pre-registration 17) **27**

27 Larger groups: an interim comparison with longer training **29**

28 Allocating rollouts by frontier value (Pre-registration 18) **30**

29 External versus self-generated information (Pre-registration 19) and the answer budget (Checks 16 and 21) **31**

30 Early reference checkpoints (Check 20) **31**

31 Moving to AWS, and a faster evaluator **31**

32 A larger model: Qwen2.5-Math-7B at 2,048 tokens (Pre-registration 20) **32**

33 A larger answer budget: 1.5B at 1,024 tokens (Pre-registration 21) **32**

34 Guidance with human solutions inside RL (Pre-registration 22) **33**

35 Minimum-help credit (Pre-registration 23, pilot; 24 and 25) **33**

36 Do methods move along one shared direction? (Pre-registration 26) **33**

37 Same problems, same rates: where the gain comes from (Investigation 27, exploratory) **35**

38 Comparison with the budgeted-search law (restricted) **35**

39 Running now **35**

40 Where we stand **36**

A Methods reference **37**

B Experiment tracker (Pre-registrations 12–28) **38**

C Compute and engineering **40**

1 The problem in plain words

Large language models can now solve competition math problems. A big part of how they got there is a training method called **reinforcement learning with verifiable rewards (RLVR)**: the model writes an answer to a math question, a simple program checks whether the final number is right, and the model is nudged to make answers like the right ones more likely and answers like the wrong ones less likely. Repeat this over thousands of questions and the model improves. This recipe is behind DeepSeek-R1 and many open models.

The loop has many design choices that look small but are not:

- **How many answers to write per question.** Writing answers is most of the cost of training.
- **Which answers to push up or down, and by how much.** For example, reward only the correct answers, or only penalize the wrong ones, or both.
- **What to aim for.** Getting one answer right ("pass@1") or getting at least one right out of several tries ("pass@k", a measure of diversity).

Every one of these choices changes three things at once: *which direction* the model is pushed, *how noisy* that push is, and *how much it costs*. Papers usually compare such choices by trial and error, often with unequal compute, one random seed, and small test sets. This project set out to make the effects exact and measurable, and then to test them carefully.

The headline result so far: **rewarding correct answers and penalizing incorrect answers push the model in the same direction on average; what separates them is noise, and incorrect answers carry most of it.** A well-known paper (Zhu et al., NeurIPS 2025) argued that penalizing incorrect answers is special because it preserves diversity. In our experiments it behaves like a noisier, slower version of rewarding correct answers. Whether that holds for another model family, and whether the noise is truly the *cause*, are the two experiments running now.

2 Background you need

The model as a "policy"

A language model given a question x defines a probability $\pi(y \mid x)$ for every possible answer y (a sequence of tokens). Sampling from it at temperature 1 gives a random answer. In RL language the model is a *policy*: it chooses what to say.

Reward. A verifier compares the answer's final boxed number with the known solution: reward $r = 1$ if right, $r = 0$ if wrong. For a question, the model's **success probability** p is the chance that a sampled answer is right.

pass@1 and pass@k. pass@1 is the average success probability over test questions (accuracy). pass@k is the fraction of questions where at least one of k sampled answers is right. With n samples of which c are right, the unbiased estimate is $1 - \binom{n-c}{k} / \binom{n}{k}$. pass@$k$ at large k measures *coverage* or *diversity*: whether the model can still find the correct answer somewhere among many tries. Training usually raises pass@1 and can lower pass@k at large k, because the model becomes confident and repetitive.

Policy gradient. To increase the expected reward, move the model's parameters θ along its gradient. The classic identity (REINFORCE, Williams 1992) is

$$\nabla_\theta \, \mathbb{E}[r] = \mathbb{E}\big[r \, \nabla_\theta \log \pi(y \mid x)\big].$$

The vector $\ell(y) = \nabla \log \pi(y)$ is the answer's **score**: the direction that makes that answer more likely. So the training update is a sum over sampled answers, $g = \sum_i A_i \, \ell(y_i)$, where A_i is the answer's **advantage**: positive pushes the answer up, negative pushes it down.

Baselines. Because $\mathbb{E}[\ell(y)] = 0$ (probabilities sum to one), subtracting a constant b from every reward, $A = r - b$, does not change the *average* update. It does change its *noise*. This is classical (Williams 1992; Greensmith, Bartlett and Baxter 2004) and matters a lot later.

GRPO and Dr. GRPO. The standard recipe samples a *group* of n answers per question (here $n = 8$) and uses the group's mean reward as the baseline. GRPO (DeepSeekMath, 2024) also divides by the group's standard deviation and averages each answer's loss over its tokens; Dr. GRPO ("GRPO done right", Liu et al. 2025) showed those two divisions bias training and removed them, giving $A_i = r_i - \bar{r}$. Example: if 2 of 8 answers are right, correct answers get $+0.75$ and wrong ones -0.25. Dr. GRPO is our main baseline.

PSR and NSR. Two extreme choices studied by Zhu et al. (2025): *positive sample reinforcement* (PSR) gives $A = 1$ to correct answers and 0 to wrong ones (only reward successes); *negative sample reinforcement* (NSR) gives $A = -1$ to incorrect answers and 0 to right ones (only penalize failures). W-REINFORCE, their recommendation, gives 0.1 to right and -1 to wrong.

Signal and noise. Each update is a random vector: its average (over the random answers) is the *signal*, the direction you want; its fluctuation around that average is the *noise*. A noisy update wastes part of each step. In ordinary deep learning, the *gradient noise scale* (McCandlish et al. 2018) says how many samples per step are needed before noise stops mattering.

Adam. The optimizer used everywhere. It keeps running averages of each coordinate's gradient ("momentum") and squared gradient, and divides the first by the square root of the second. A consequence that matters here: more noise in the gradient inflates the denominator, so Adam takes *smaller effective steps* along the signal. Noise therefore slows Adam down.

On-policy. We always sample fresh answers from the current model and update once on them. Many systems (for example the verl framework) reuse each batch of answers for several updates, with a "clipped" loss that limits how far the model moves from the one that wrote them.

Pre-registration. Before each experiment we wrote down, in a dated file, the design, the prediction and the rule that decides the verdict, so that results could not quietly reshape the question. All files are in `experiments/training/PREREGISTRATION_*.md`.

3 The literature, and where the gaps are

3.1 RLVR recipes

GRPO (Shao et al., DeepSeekMath, 2024) made group-based RL standard for math; DeepSeek-R1 scaled it. Dr. GRPO (Liu et al. 2025) removed GRPO's length and standard-deviation biases. DAPO (Yu et al. 2025) added "dynamic sampling": drop questions whose 8 answers are all right or all wrong (they carry no learning signal under a group-mean baseline) and refill the batch.

3.2 Spending compute on answers: adaptive rollout collection

Since writing answers dominates cost, many 2025–2026 methods decide adaptively how many answers each question gets: probe a few answers and continue only on questions with mixed results (SPEED-RL, SARA, Pilot-Commit, GRESO), allocate a budget across questions (Knapsack RL, Reinforce-Ada, PAIR), or sample until a correct answer appears. Our review of these papers found recurring evaluation flaws:

- gains plotted against training *steps* while spending 2 to 3 times more answers per step (for example, SARA's +6.8 points comes with 3.3 times the answers);
- learning rates not retuned, single seeds, 30-problem benchmarks, best checkpoints chosen on the test set;
- the claimed mechanism ("lower variance") asserted but never measured.

3.3 Theory of the update

Davis and Recht (2025) and ARCUS (2026) derive the *average* update of fixed-size groups: on average, a group's update points along ∇p, scaled by a function of p. McCandlish et al. (2018) introduced the gradient noise scale; Bansal, Mohri, Abreu, Alvarez-Melis and Kakade ("Critical Batch Size for LLM Policy Optimization", HiLD workshop at ICML 2026) apply it to GRPO, split the noise into a between-question and a within-question part, and find the within-question part dominant. Li et al. (2026) study square-root learning-rate scaling; IsoCompute (2026) studies answers per question; Malladi et al. (2022) explain square-root scaling for Adam.

3.4 Assumptions about noise

Recent variance-reduction papers model the noise with simplifying assumptions. VSR (Wang, Liu and Wang, October 2026) states as Assumption B.3 that the expected squared score norm does not depend on whether the answer was right; BICC (2026) uses a similar approximation. Nobody, to our knowledge, had measured whether right and incorrect answers have different gradient sizes.

3.5 Does RL add reasoning, or only sharpen it?

Yue et al. (2025) showed that RL-trained models beat their base models at pass@1 but can lose at large k: RL may sharpen what the base model already does rather than add new abilities. Wen et al. (2025) introduced **CoT-Pass@k**, which counts an answer only if its reasoning is also valid, and argued RL does improve reasoning under that measure. "Spurious Rewards" (Shao et al. 2025) found that Qwen2.5-Math improves even with random rewards, so small effects on that model family are suspect.

3.6 Positive versus negative reinforcement

Zhu et al. ("The Surprising Effectiveness of Negative Reinforcement in LLM Reasoning", NeurIPS 2025) trained with only negative samples (NSR) or only positive samples (PSR). They report that NSR preserves entropy and pass@k while PSR collapses them, credit the negative samples (pushing down incorrect answers redistributes probability to other candidates), and recommend W-REINFORCE. They used 4B to 8B models and k up to 256.

3.7 Objectives aimed at pass@k

PKPO (Walder and Karkhanis 2025) and advantage-shaping work (Thrampoulidis et al. 2025) give unbiased gradient estimators for pass@k, so a model can be trained directly for diversity.

3.8 The gaps

Gap in the literature	What this project does about it
The *noise* of adaptive answer-collection rules (random group sizes, early stopping) is not characterized; only the average is.	An exact two-moment theorem: average and noise of any such rule from four numbers per question (Section 5).
Nobody measures gradient size separately for right and incorrect answers; recent papers assume they are equal.	We measure it: incorrect answers carry 2.4 to 3.1 times larger squared gradients (Section 6).
Adaptive-rollout gains are compared at unequal compute, with one seed.	All our comparisons are at matched generated answers, paired per problem, pre-registered, with at least two seeds.
The NSR explanation (negative samples preserve diversity) was never separated from the fact that NSR and PSR differ in noise.	We show their average updates are identical, measure their noise, and compare diversity at *matched accuracy* (Sections 9–12).
"Does RL preserve diversity?" is usually answered at a fixed step, which confuses slower learning with preserved diversity.	We compare runs at equal pass@1: an accuracy-versus-diversity curve.

4 The earlier idea (STOP) and why it failed

Why we did it

Before this project, the plan was STOP: a better way to give credit inside an answer. Segment-level RL splits each answer into blocks and credits every block with the change in estimated success across it. STOP asked *where the blocks should end*. Existing methods pick cut points with local scores (token entropy, value variance, EPIG-Tree's gradient-aware score). We showed with exact small examples that a local score can pick the wrong cut, because a cut changes its neighbors and because token gradients interact, and planned a geometry-aware cut rule.

The theory worked. The update error splits exactly into a part that ignores gradient geometry and an interaction part; the interaction vanishes when token gradients are orthogonal; a cut's benefit depends on the other cuts only through its two neighbors.

What we found

In Qwen2.5-Math-1.5B, gradients at different positions are almost orthogonal (median $|\cos| \leq 0.005$) and the small interactions cancel in aggregate (about $\pm 1\%$). In a pre-registered test over all partitions of a 64-token grid, interactions reversed only 2.2% of orderings, none statistically clear, and never changed the best single cut; the geometry-free score lost at most 1.3% of the available gain. The effect STOP was built to exploit is real in constructed examples but too small in a real model to matter. Written up honestly as "local scores suffice" (TMLR or workshop level).

What we kept: exact per-answer gradient measurement tools, and the habit of *measuring the mechanism before building a method*. Both made the next project possible.

5 What is each generated answer worth? (WORTH)

Why we did it

Generating answers is the main cost of RLVR, and the adaptive-collection literature (Section 3) compares rules by heuristics. If the value of each answer could be computed exactly, rules could be ranked, and the best rule derived instead of guessed.

5.1 The two-moment theorem

Let p be a question's success probability, F_1 and F_0 the average squared score size $\mathbb{E}\|\ell\|^2$ of right and of incorrect answers, and $\|\nabla p\|^2$ the size of the question's learning signal. Two simple facts drive everything: a correct answer's average score is $\nabla p/p$ and a incorrect answer's is $-\nabla p/(1-p)$; and given its reward, an answer is independent of the others.

Theorem (two moments of any reward-adapted rule)

For any rule that decides when to stop sampling and which advantages to give using only the rewards seen so far (including random group sizes, early abandonment, sampling until success, and filtering),

$$\mathbb{E}g = c(p)\,\nabla p, \qquad \mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p),$$

where c, D_1, D_0, X and the expected cost depend only on the rule and on p. So the average, the noise and the cost of the update depend on a question only through **four numbers**: p, $\|\nabla p\|^2$, F_0, F_1.

Why it is true, in one line: condition on the whole sequence of rewards; then the stopping time and the advantages are fixed numbers and the answers are independent, so averaging over answers and then over rewards gives both formulas. No optional-stopping argument is needed. **What is new** is the noise part with separate F_0, F_1 and random group sizes; the average part for fixed groups was known (Davis and Recht; ARCUS). All statements are verified by exact enumeration in small models (`theory/exact.py`).

5.2 Consequences: ranking the rules and the optimal rule

Treating questions as having unrelated gradients and choosing the step size optimally, the progress per generated answer is proportional to signal^2/(cost $\times$ second moment). Dynamic programming over the counts of (wrong, right) answers seen so far gives the **optimal rule**: after each block of 4 answers, a lookup table says stop or continue, and which advantages to give.

Rule (efficiency relative to the optimal rule)	Uniform	Mostly hard	U-shaped	Mostly easy
Fixed group of 8 (GRPO, Dr. GRPO, DAPO)	0.61	0.38	0.40	0.38
Fixed group of 16	0.65	0.41	0.43	0.41
Probe 4, abandon if all agree, fill to 16	0.77	0.60	0.67	0.60
Sample until 1 correct, cap 16	0.26	0.09	0.12	0.30

Table 1: Exact efficiency by difficulty mix. DAPO's filtering has exactly the efficiency of the group it filters; sample-until-correct is worst; probe-then-fill is the best rule in use.

5.3 When does a better rule matter? The critical budget

Without assuming unrelated question gradients, one step's progress at B answers per step is $\langle \nabla J, \mu \rangle^2 / (2L(s + nV/B))$, with s the squared size of the mean update and V its noise. The **critical budget** $B^\star = nV/s$ splits two regimes: below it, a rule k times more efficient is worth k times the answers (about $\sqrt{k}$ under Adam at a fixed learning rate); above it, no reweighting helps (Figure 1). This later turned out to be already published for GRPO (Bansal et al.), so it became background.

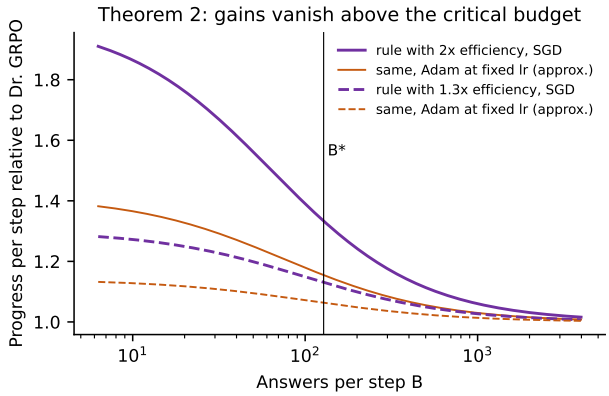


Figure 1: Predicted gain of a better rule against answers per step: gains vanish above the critical budget, and Adam turns a gain of k into about $\sqrt{k}$.

6 Measuring the four numbers

Why we did it

The theory's predictions depend on two unknowns: how F_0 compares with F_1, and how the learning signal $\|\nabla p\|^2$ varies with difficulty. Both had to be measured in a real model before any prediction meant anything.

What we found

Incorrect answers carry larger gradients. In Qwen2.5-Math-1.5B (exact per-answer gradients on MATH training problems): in the last-layer block, 40 questions by 16 answers, $F_1/F_0 = 0.36$ (90% intervals do not overlap), while incorrect answers are only 9% longer (421 against 386 tokens). Over all 1.5B parameters, measured before every training run: F_1/F_0 between 0.32 and 0.42, so **incorrect answers carry 2.4 to 3.1 times larger squared gradients**. Likely reason: incorrect answers contain more low-probability tokens, whose scores are large. This contradicts VSR's Assumption B.3.

The learning signal fades at both ends. With 96 answers per question, the estimated $\|\nabla p\|^2$ is 0.73 ± 0.60 on very easy or very hard questions and 15.5 ± 5.4 on medium ones; the best fit is $p^2(1-p)^2$, and a flat shape is rejected (Figure 2). This makes the optimal rule robust: it should spend answers on medium questions.

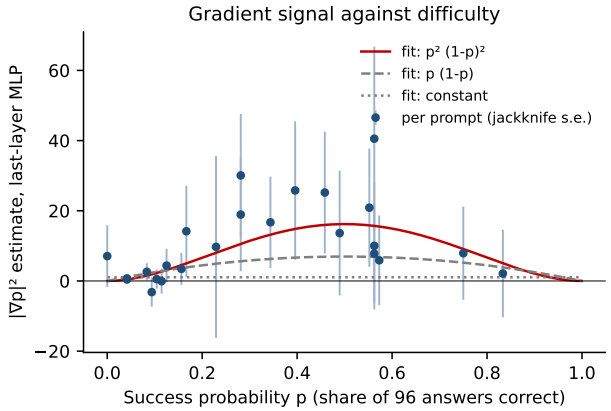


Figure 2: Per-question learning signal against success probability, with three candidate shapes.

7 First tests of the theory

7.1 Pre-registration 1: scale down incorrect answers (pilot)

Why we did it

The simplest consequence of $F_0 > F_1$: weight each incorrect answer's advantage by $w = F_1/F_0$. A one-line change to Dr. GRPO, predicted to raise efficiency by 1.1 to 1.6 times in 11 of 12 settings.

Design. Qwen2.5-Math-1.5B base, 3,920 numeric MATH training problems (four subjects), 16 questions by 8 answers per step, 150 steps, 512-token answers, AdamW at 10^{-6}; Dr. GRPO against the scaled version; two seeds; primary: greedy accuracy on the 364 numeric MATH-500 problems. **Prediction:** promising if at least +2 points in both seeds.

What we found

No gain: 45.3% against 47.5% (seed 1), 44.0% against 46.7% (seed 2). Verdict: **no evidence**. Lesson: under Adam at a fixed learning rate, a 1.1 to 1.6 times efficiency gain shrinks to 1.06 to 1.25 times, too small to see with two seeds.

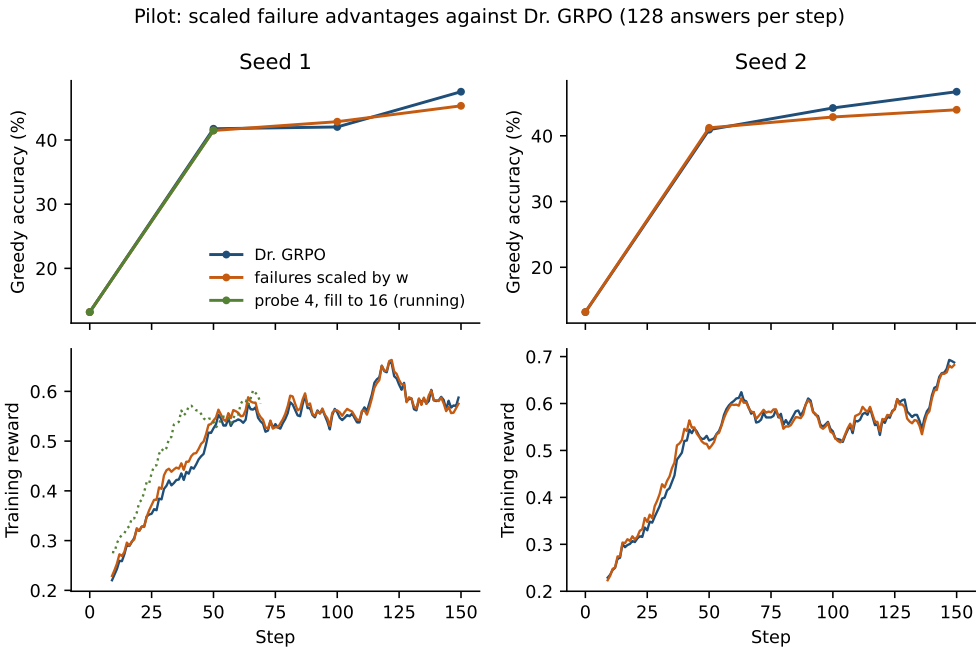


Figure 3: Pilot: scaled failures (orange) track Dr. GRPO (blue).

7.2 Pre-registrations 2 and 3: collection rules at equal cost, and measuring noise inside training

Why we did it

The optimal rule predicts a larger gain (1.7 to 2.3 times Dr. GRPO's efficiency). Testing it fairly needs exactly matched generated answers. And to know *why* a rule helps or not, we needed to measure signal and noise during training, not just accuracy.

Design. The optimal rule (lookup table re-solved every 10 steps from the recent difficulty mix), probe-then-fill, scaled failures, and Dr. GRPO at several budgets (32 to 256 answers per step), all at matched answers (19,200 over 150 steps for the 128-answer arms), two seeds, paired per problem on 3,382 held-out MATH test problems. **New tool:** at every step the questions are split at random into two halves; inner products of the two halves' gradients, and of each half with an independent unbiased reference for the population gradient, give the signal, the noise and the noise scale, without extra sampling. **Prediction:** promising if the optimal rule beats Dr. GRPO by at least 1.5 points in both seeds.

What we found

- Optimal rule: $+1.1 \pm 0.4$ (seed 1) and $+3.3 \pm 0.5$ (seed 2) points, with matched answers (19,172 and 19,200) and about 1% more tokens. Positive in both seeds, but the pre-registered bar (1.5 in both) was **missed** because of seed 1.
- Probe-then-fill: +2.0 and +0.8 (MATH-500). Scaled failures: −0.4 and +1.1. Fewer answers clearly hurts: 32 against 128 answers per step, −4.4 points (seed 1; not every rung of the budget ladder obtained GPUs).
- Training becomes noise-limited as it plateaus: Dr. GRPO's noise scale grows from about 900 answers (steps 0 to 49) to over 10,000 (steps 50 to 149): the shared signal shrinks while the noise does not.
- Gains are much smaller than one-step theory predicts, consistent with Adam's square-root effect.

8 The literature check and the change of direction

Why we did it

Before investing more, we checked carefully (three literature reviews, key claims verified by hand) whether the framing was new.

What we found

- **The critical-budget framing was taken** (Bansal et al., HiLD at ICML 2026), including the finding that within-question noise dominates.
- The common evaluation flaws listed in Section 3.
- **No paper measures gradient norms by correctness**, as far as we found, and none separates the average and the noise of positive and negative reinforcement.

So the main question moved to positive versus negative reinforcement, where our theorem and our measurement give something nobody else has. The optimal-rule result stays as a practical side result.

8.1 The key observation

Give each correct answer advantage a and each incorrect answer b. The theorem gives

$$\mathbb{E}g = n(a-b)\,\nabla p, \qquad \text{noise} \approx n\big(a^2 p F_1 + b^2(1-p)F_0\big).$$

- PSR ($a = 1, b = 0$) and NSR ($a = 0, b = -1$) both have average $n\nabla p$: **exactly the same expected update**. (This is the classical baseline invariance; the new part is the noise and its measurement.)
- In fact NSR = PSR $- \sum_i \nabla \log \pi(y_i)$, and the extra term has average exactly zero: it is pure noise.

- PSR's noise npF_1 sits on easy questions; NSR's $n(1-p)F_0$ on hard ones. With $F_0 \approx 2.5F_1$, NSR is noisier on every question solved less than 71% of the time.
- The whole family $A = r - b$ shares the average; its noise is smallest near $b^\star = pF_1/(pF_1 + (1-p)F_0) \approx 0.15$.

Consequence: to first order, any difference between PSR and NSR (entropy, pass@k) must come from the noise, from an optimizer that reacts to noise (Adam), or from implementation choices outside the theorem.

9 PSR against NSR (Pre-registration 4)

Why we did it

Test the observation directly: if NSR's famous diversity advantage is real in clean on-policy training with equal averages, it must be a noise effect; if it does not appear, it must come from implementation.

Design. Dr. GRPO, PSR, NSR, W-REINFORCE; two seeds; strictly on-policy; in-training noise measurement; an on-policy entropy probe (the first four answers of every question, identical for all rules); pass@k with 64 samples at step 150. **Prediction:** separation if NSR beats PSR by at least 3 points of pass@64, or has at least 20% more entropy, in both seeds.

Rule (seed 1 / seed 2)	**Training acc., steps 25–49**	**pass@1 at 150**	**pass@64 at 150**	**Entropy, 100–150**
Dr. GRPO	50.6 / 48.0	43.3 / 43.2	83.2 / 81.9	0.227 / 0.224
PSR (reward right only)	51.3 / 48.2	42.4 / 42.0	83.2 / 83.8	0.227 / 0.226
NSR (penalize wrong only)	33.6 / 36.4	37.7 / 39.1	83.0 / 82.1	0.251 / 0.251
W-REINFORCE	38.6 / 38.0	38.5 / 39.8	80.5 / 81.3	0.243 / 0.256

Table 2: Untrained model: pass@1 13.8, pass@64 79.9.

What we found

NSR's measured noise is 6.4 times PSR's (theory about 7). NSR learns clearly more slowly, ends with lower pass@1, and reaches the *same* pass@64; its entropy is only 11% higher. The measured noise matches the theorem within 30% for every rule. Verdict: **the separation is not reproduced** in either seed. W-REINFORCE, the NSR paper's recommendation, has the lowest pass@64.

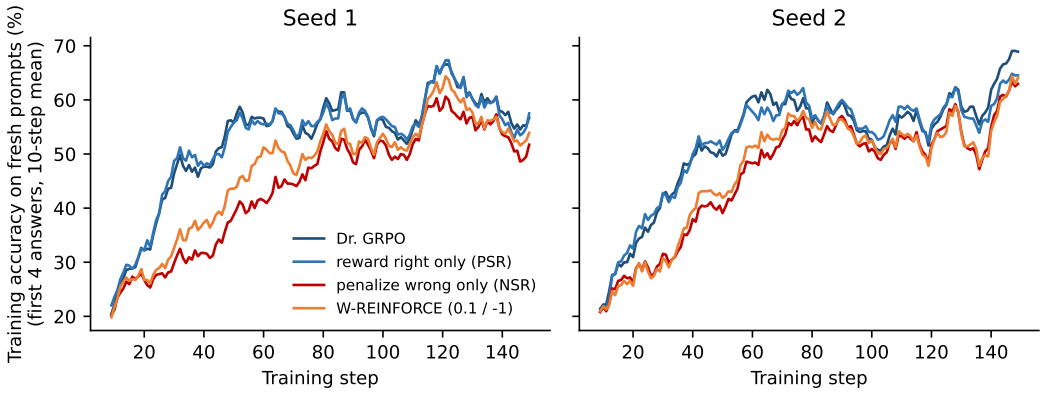


Figure 4: Training accuracy on fresh questions. The noisy rules, dominated by incorrect answers, learn about twice as slowly.

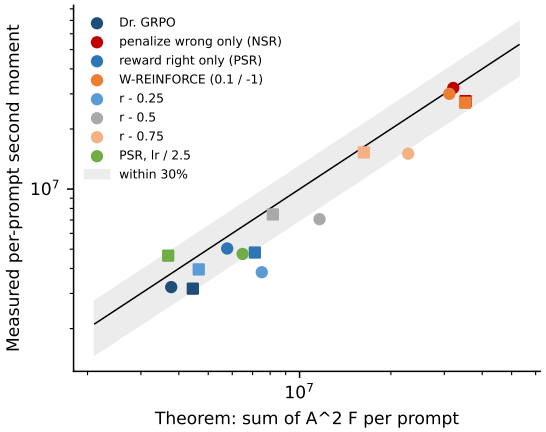


Figure 5: Measured gradient noise against the theorem's prediction from F_0, F_1 and the observed reward counts; shaded: within 30%.

10 Matched accuracy, and the NSR paper's own implementation

10.1 Pre-registration 5: is NSR just a slower PSR?

Why we did it

At a fixed step, a slower run looks more diverse simply because it has learned less. The fair comparison is diversity *at the same accuracy*: for each NSR checkpoint, interpolate PSR's pass@32 at the same pass@1 along PSR's trajectory and take the difference. This gives an **accuracy-versus-diversity curve** per run.

Design. PSR and NSR, seeds 3 and 4, pass@32 from 32 samples on 128 MATH-500 problems at steps 0 to 150. **Prediction:** the matched difference is below 2 points in both seeds.

What we found

NSR is 2.1 points *below* PSR's curve (seed 3) and 0.2 below (seed 4). **No extra diversity at equal accuracy** (Figure 6, left).

10.2 Pre-registration 6: do the NSR paper's implementation choices create the effect?

Why we did it

Zhu et al. used verl, which differs from clean training in three ways the theorem does not cover: each batch of answers is reused for 4 updates with clipping (stale samples), each answer's loss is averaged over its tokens, and PSR and NSR average over only the answers they train on. If the effect lives anywhere, it may live there.

Design. All three choices reproduced at matched answers and updates (with fresh samples, our clipped loss equals the plain loss exactly). **Prediction:** recreated if NSR beats PSR by 3 points of pass@32, or 20% entropy, in both seeds.

What we found

NSR learns even more slowly here (training accuracy 31.0 and 31.4 against 43.0 and 45.4), and at step 150 its pass@32 is 7.0 and 1.6 points *lower* than PSR's; its entropy is 1.11 and 1.18 times PSR's; at matched pass@1 the differences are +0.3 and −1.3. Verdict: **not recreated**. Disclosure found later: per-token averaging makes gradients larger, so the gradient clipping at norm 1 (verl's default) was active on 26 to 28% of NSR's steps and 7 to 8% of PSR's; it never triggered in any other on-policy run.

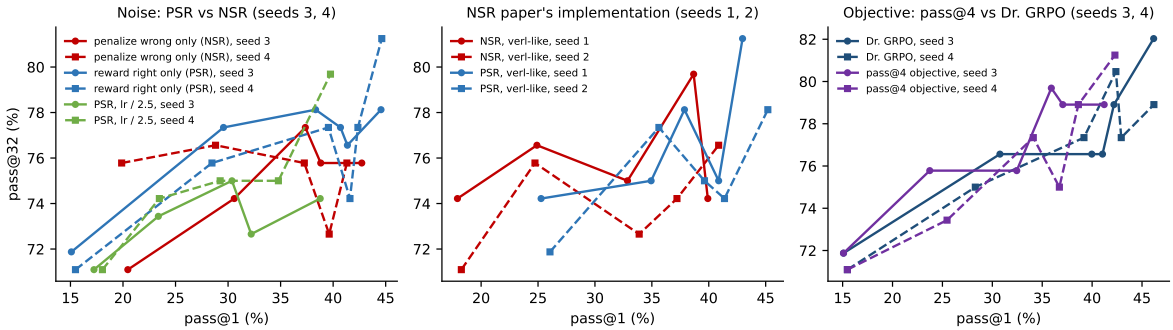


Figure 6: Accuracy (pass@1) against diversity (pass@32) during training, 128 problems. Left: changing the noise (PSR, NSR, PSR at a lower learning rate). Middle: the NSR paper's implementation. Right: changing the objective (pass@4 against Dr. GRPO). No run separates from its reference beyond the evaluation noise (point comparisons, not yet an equivalence test).

11 Same average, graded noise (Pre-registration 7)

Why we did it

The strongest test of "only noise differs" available with one family: the baselines $A = r - b$ for $b = 0, 0.25, 0.5, 0.75, 1$ all have exactly the same average update, but the theory predicts different noise. If learning speed follows the predicted noise, noise is the likely cause. A PSR run with the learning rate divided by 2.5 tests the "slower clock" reading: does NSR look like PSR run more slowly?

	$b = 0.25$	$b = 0$ (PSR)	$b = 0.5$	$b = 0.75$	$b = 1$ (NSR)	PSR, lr /2.5
Predicted noise vs PSR	0.81×	1×	1.25×	2.5×	about 6×	1×
Steps to 45% training accuracy, seed 3	37	44	44	53	56	84
Steps to 45% training accuracy, seed 4	38	33	41	45	56	76
pass@32 vs PSR at equal pass@1, seed 3	+0.0	0	+0.8	−0.8	−2.1	
pass@32 vs PSR at equal pass@1, seed 4	+0.2	0	+2.1	+0.6	−0.2	

What we found

- **Speed follows predicted noise** in both seeds (rank correlation 0.9 to 0.97; pre-registered threshold 0.8): passed.
- The noise formula is within 30% in all seed-4 runs but only 0.51 to 0.73 in seed 3, so 12 of 16 runs overall; likely because gradient sizes are measured once, before training, and shrink as the model sharpens.
- Every run stays within about ±2 points of PSR's curve; "noise adds exploration" is rejected. The strict slower-clock rule asked for every difference below 2 and two were 2.07 and 2.14: **formally inconclusive**. PSR at a 2.5 times lower learning rate learns *more* slowly than NSR, so NSR equals a smaller slowdown.

12 Changing the objective, and grading the reasoning (Pre-registration 8)

Why we did it

If noise only changes speed, perhaps changing *what the update aims at* moves the curve. We aimed the update exactly at pass@4: per-answer weights from the unbiased pass@k estimator with a leave-one-out baseline, verified over all 256 reward patterns to average $2 \cdot 4(1-p)^3 \nabla p$, the pass@4 gradient (such objectives are known, e.g. PKPO; here it is a control). It pushes hard questions and leaves solved ones alone. We also graded the reasoning, not just the final number, because a diversity gain could hide lucky guesses.

Prediction: the curve moves if pass@4 training is at least 2 points of pass@32 above Dr. GRPO at equal pass@1 in both seeds.

What we found

+1.6 (seed 3) and +0.2 (seed 4): **curve fixed**. At step 150 pass@4 training simply has lower pass@1 (41 to 42 against 46). With reasoning graded by Claude Haiku 5.5 (three judgments per correct answer, majority vote; cost about $9):

Step 150, seed 3 / seed 4	pass@1	CoT-Pass@1	pass@32	CoT-Pass@32
Dr. GRPO	46.2 / 46.2	43.4 / 43.3	82.0 / 78.9	74.2 / 70.3
pass@4 objective	41.2 / 42.3	38.6 / 39.3	78.9 / 81.2	72.7 / 72.7
Untrained model	15.1	13.5	71.9	66.4

Reasoning-checked diversity is the same (CoT-Pass@32 72.7 against 72.3 on average); about 93% of trained models' correct answers have valid reasoning, up from 90%.

13 What the Qwen experiments say together

The picture after eight pre-registrations

In on-policy RLVR on Qwen2.5-Math-1.5B, every reward-shaping choice we tried (PSR, NSR, five baselines in between, the NSR paper's implementation, and a pass@4 objective) moved the model *along* one accuracy-versus-diversity curve at different speeds; none moved the curve beyond about 2 points. The speed followed the noise that the theorem predicts from four numbers per question. NSR's "preserved diversity" is what any slower run looks like at a fixed step.

Two side results: the theory explains why standard practice is close to right (Dr. GRPO's per-question baseline is 1.25 times less noisy than the best possible constant baseline), and the optimal answer-collection rule moves faster along the curve at equal cost. A planned method, WORTH-k (a noise-minimized pass@k objective), was dropped: it assumed that changing the objective moves the curve, and Pre-registration 8 found it does not at this scale.

14 An independent review, and what we changed

Why we did it

The goal is an ICML paper. Before spending more, we asked an independent model (Codex) to review the project as a demanding ICML reviewer and area chair, and since then it reviews every experiment before launch.

What we found

Verdict: **reject as it stands, 4 out of 10 (about 10 to 20% acceptance), with a credible path to a main-track paper**; about 6 out of 10 (30 to 45%) if a second model family establishes one curve with intervals; 5 out of 10 if the curves robustly differ. Its main points, all accepted:

1. **Novelty is narrower than framed.** Equal averages for PSR and NSR are classical baseline invariance; pass@k objectives exist (PKPO); Dr. GRPO, the coverage framing (Yue et al.) and CoT-Pass@k (Wen et al.) are prior work. Ours: the noise accounting with $F_0 \neq F_1$ and adaptive group sizes, the measured asymmetry, and the matched-accuracy tests.
2. **"No difference detected" was presented as "the same".** One 1.5B model, two seeds per comparison, 128 problems, point estimates; the NSR paper used 4B to 8B models and k up to 256.
3. **Noise predicts speed, but cause is not shown.** The baseline family changes the noise and the push on incorrect answers together.
4. Four numbers give the *size* of the noise, not its geometry or what Adam does with it.

What we changed (no GPU needed): the report, the collaborator brief and the story now credit baseline invariance and PKPO, drop every "nobody measured this" claim, describe the Qwen result as absence of detected separation, state the scale gap to the NSR paper, and disclose the clipping in Pre-registration 6. **What we launched:** the two experiments below, each reviewed by Codex several times (Pre-registration 9 four times, Pre-registration 10 three times) and changed after every review.

15 A second model family (Pre-registration 9)

Why we did it

Qwen2.5-Math reacts even to random rewards, so small effects on it are suspect. The claim had to hold on a different family, for longer training, with a larger evaluation and proper equivalence intervals rather than "no difference seen".

Design. OLMo-2-1B-SFT (AllenAI, never RL-trained), all seven MATH training subjects (5,685 problems), 300 steps, answers up to 1,024 tokens, five rules (Dr. GRPO, PSR, NSR, the pass@4 objective, the optimal collection rule) and two seeds; checkpoints every 50 steps evaluated with 32 samples on 256 held-out, subject-stratified problems; greedy accuracy on 3,382 test problems at the end.

What we found

Test	Result	Verdict
Incorrect answers have larger gradients	2.14 [1.55, 2.93] and 2.54 [1.73, 3.65]	**passed**, both seeds
NSR noisier than PSR	19.5× and 22.2×	**passed**
NSR slower to reach the midpoint	seed 1 tie (52 vs 52), seed 2 88 vs 41	failed (needs both)
NSR vs PSR, same curve (primary)	−0.30, 90% [−1.28, +0.77]	**equivalent within** ±2
pass@4 objective vs Dr. GRPO	+1.32, 90% [+0.24, +2.26]	inconclusive (a small lift)
Optimal rule vs Dr. GRPO, equal answers	+0.65 [−0.24, +1.54]	inconclusive

The main claim replicates on a second family: penalizing incorrect answers lies on the same accuracy–diversity curve as rewarding right ones. No formatting flag; removing the one problem whose answer key was wrong changes nothing.

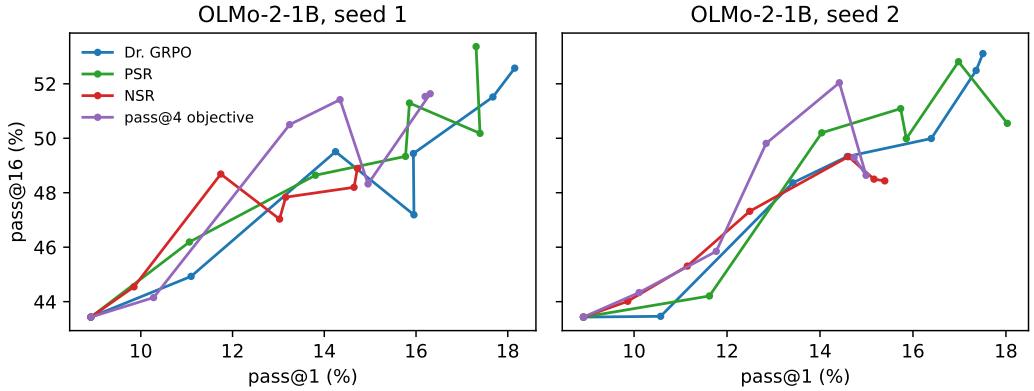


Figure 7: OLMo: accuracy (pass@1) against diversity (pass@16) at steps 0 to 300. The rules lie on one curve.

16 Is the noise the cause? (Pre-registration 10)

Why we did it

The reviewer's main objection: noise *predicts* learning speed, but in the baseline family more noise always came with a harder push on incorrect answers, so the cause was not shown.

The intervention. Add to PSR a random-sign term whose size, at every step, matches NSR's predicted noise. It averages to zero, so the expected update is unchanged, and it never pushes incorrect answers down systematically. If this "PSR + noise" learns as slowly as NSR, noise of that size is enough to cause NSR's lag. Three fresh seeds; also NSR and PSR at twice the learning rate; an NSR "shadow" computed on the same answers to check the noise really matches.

What we found

- The noise matched: 1.02 [1.01, 1.05] times NSR's.
- **PSR + noise is as slow as NSR**: speed $\phi = -0.11$ $[-0.22, -0.05]$ on a scale where NSR is 0 and PSR is 1 (pre-registered margin $\pm 1/3$; passed; one seed flagged).
- NSR at twice the learning rate recovers its whole lag ($\phi = 1.16$); at a matched doubled rate the gap shrinks to 31%.
- **Same curve**, now with intervals on Qwen: NSR vs PSR -0.20 $[-0.97, +0.55]$, PSR + noise -0.37 $[-1.13, +0.46]$, NSR at $2\times$ -0.03 (all equivalent within ± 2).
- The noise formula fits training within 0.69–1.18 once gradient sizes are re-measured (the asymmetry fades from about $3\times$ to $1.5\times$).
- Adam: NSR's raw noise is $6\times$ PSR's but only 1.0–1.3$\times$ after Adam's scaling; NSR's steps point 13–27% less along the true improvement direction. Noise does not shrink Adam's steps, it misdirects them.

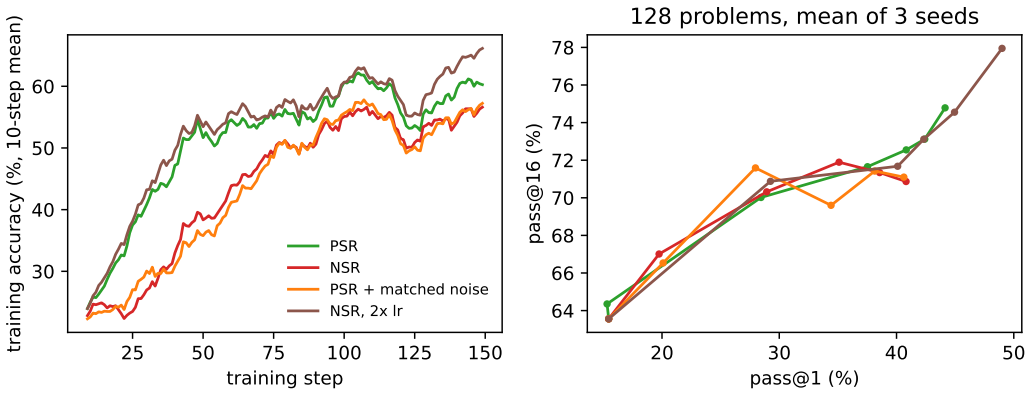


Figure 8: Left: training accuracy; PSR + matched noise (orange) learns as slowly as NSR (red); NSR at twice the learning rate (brown) as fast as PSR (green). Right: all four lie on one accuracy–diversity curve.

17 Is the curve a temperature effect? (Pre-registration 11)

Why we did it

By our proposition, diversity at matched accuracy is how evenly success is spread over questions. If RL simply made the model globally more confident (like lowering the temperature), then sampling a late checkpoint at a higher temperature should beat the training curve, and a method that limits sharpening would have room to work. This decides which method to build.

Design. The PSR checkpoints of Pre-registration 10 and the base model, sampled at temperatures 0.5 to 1.6; the quantity D is the late checkpoint's pass@16 at a raised temperature minus the training trajectory's pass@16 at the same pass@1. The model's entropy is also measured on fixed

base-model answers.

What we found

- **RL does not sharpen globally.** On fixed text the entropy barely changes (1.081 for the base, 1.064 after training), while on the model's own samples it falls three- to fourfold: RL steers into confident paths.
- **Undoing sharpening does not help.** At the temperature that restores the base entropy ($T^* = 1.01$), $D = 0.0$ in every seed; slightly higher temperatures stay on the curve; at 1.3 formatting breaks and coverage falls. Pre-registered verdict: inconclusive ($D = -1.54$ on 256 problems; -1.42, 90% $[-2.19, -0.68]$, on 511).
- RL is far above anything temperature does for the base model (+7 to +16 points): RL adds ability, not only sharpening.

Consequence (declared in advance): the memory branch of the method experiment runs.

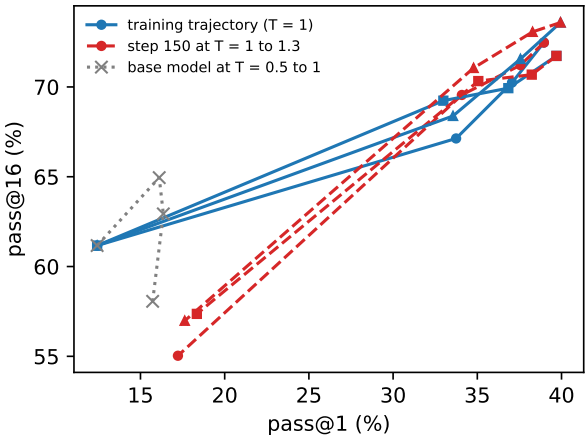


Figure 9: Training trajectory (blue), the late checkpoint at higher temperatures (red), and the base model at lower temperatures (gray).

18 Checking the answer keys

Why we did it

An independent review found two wrong reference answers in the evaluation sets. Before using them to judge a method, every answer had to be checked.

What we found

Claude Haiku 5.5 solved all 768 evaluation and validation problems; where it disagreed with the key, Claude Opus 5.5 adjudicated. **3 of 768 keys were wrong (0.4%)**: a sign error (corrected) and two questions with several answers whose key kept one (excluded). They affect all methods equally; excluding the one in the OLMo set changes no verdict.

19 From diagnosis to method (Pre-registration 12)

19.1 What the diagnosis says a method must do

Diagnostic finding	Requirement for a method
Correctness-only rules (PSR, NSR, baselines, pass@k weights, adaptive sampling) only reweight questions and change noise; they share one curve	Bring in information the current samples do not contain
Diversity at matched accuracy = how evenly success is spread across questions	Raise success on rarely solved questions without hurting solved ones
On rarely solved questions a group of 8 is usually all wrong, and the signal fades as $p^2(1-p)^2$	Supply signal exactly there
RL does not sharpen globally; entropy control blocks learning	Do not regulate entropy
Extra noise slows learning	Keep Dr. GRPO's low-noise update as the base

19.2 The method: coverage-weighted success replay

Coverage-weighted memory (CM)

Train with Dr. GRPO as usual. Store every verified, completed, distinct correct answer the model finds (at most 4 per question). Each step, replay 8 stored questions by adding the gradient of $\lambda \sum_q \log \sum_{y \in S_q} \pi(y \mid q)$, choosing questions in proportion to the expected value of $k\,p(1-p)^{k-1}$ ($k = 16$, p the question's estimated success rate). This weight is exactly $\partial\,\text{pass@}k / \partial \log p$: replay goes to questions solved about one time in k, whose pass@k can still rise, not to questions already solved.

Plain replay of stored successes is known (MAPO, self-imitation, STaR, RAFT, ReST); the new parts are the derivation from the diagnosis and the coverage weighting, so uniform replay (M) is run as the ablation at the same strength. Audit arms: the published unlikeliness reward (U), entropy bonus (E), adaptive entropy control (A), and, for the other branch, an entropy floor on the first 32 tokens of each answer (EF).

19.3 The pilot (one seed, 100 steps, 255 validation problems)

Arm	pass@1	pass@16	gain at matched accuracy
Dr. GRPO, steps 0 / 50 / 100	15.1 / 35.5 / 39.3	63.9 / 71.1 / 72.5	reference
Coverage memory, $\lambda = 1$ (chosen)	35.8	73.3	+2.1
Coverage memory, $\lambda = 4$	33.1	70.2	0.0
Uniform memory, $\lambda = 1$	38.6	72.4	+0.2
Uniform memory, $\lambda = 4$	35.3	74.2	+3.2 (4 points slower; excluded by the rule)
Entropy bonus	38.9–39.6	72.6–72.7	about 0
Unlikeliness reward	38.7	71.7	about −0.5
Adaptive entropy	9.3–25.5	48.9–65.0	−2.5 or collapsed
Early-token entropy floor	12.6–12.8	56.1–58.6	collapsed (below the base model)

Reading. Memory is the only family that lifts the curve; at the same strength, coverage weighting gives +2.1 against +0.2 for uniform replay (one seed, noise about ±2). Entropy-based fixes do not

help or break learning, as the diagnosis predicted.

19.4 The main runs: results

Seeds 8, 9, 10; Dr. GRPO (B), coverage memory (CM, $\lambda = 1$), uniform memory (M) and the unlikeliness reward (U), 150 steps; evaluation on 511 held-out problems with 32 samples at steps 50, 100, 150, B also at $T = 1.15$ and 1.3; greedy accuracy on 3,382 test problems. Rule: CM *moves the curve* if its pooled mean pass@16 gain over B's trajectory-and-temperature envelope is at least +2 points, the pooled 95% interval is above 0, and the mean gain is positive in every seed.

Arm	Gain at matched pass@1	95% interval	Step-150 pass@1 / pass@16	Greedy test accuracy
Dr. GRPO (B)	0		44.1 / 74.5	47.9
Coverage memory (CM)	+1.19 (per seed +1.52, +1.44, +0.59)	$[+0.54, +1.79]$	42.1 / 75.2	47.7
Uniform memory (M)	+0.61	$[-0.04, +1.27]$	42.9 / 74.0	47.3
Unlikeliness (U)	+0.14	$[-0.32, +0.62]$	43.4 / 74.0	47.6

What we found

Registered verdict for CM: **average agreement** (90% interval $[+0.63, +1.69]$ inside ± 2). A small, consistent lift, larger than uniform replay and the published unlikeliness reward, but half the registered bar. Coverage weighting over uniform replay (+0.66) misses its +1 bar; CM's step-150 pass@1 is 2.05 points below B's; greedy accuracy differs by −0.18, 90% $[-0.83, +0.43]$. Both memory arms used the archive on 100% of steps after step 20. A possible explanation is limited discovery: a group of 8 finds a success with probability $1 - (1 - p)^8$, about 40% at $p = 1/16$ and 22% at $p = 1/32$. The replay result does not establish discovery as the bottleneck.

20 The 2026 literature, and what it changed

Why we did it

Before choosing the next method, we searched arXiv (October 21) for 2025–2026 work on the diagnosis and on methods (two systematic searches, every abstract opened, the closest papers read in full).

What we found

- **Theory, partly prior:** binary-reward RL as ascent on transforms of per-problem success (Davis and Recht, 2510.13651), weight-dependent convergence (Heckel et al., 2602.11128), pass@k advantage shaping as surrogate rewards (Thrampoulidis et al., 2510.23049), pass@k gradients as reweighted pass@1 gradients (2511.16231). Barakat et al. (2602.21189, NeurIPS 2026) show that reweighting toward hard prompts can rotate the aggregate update through shared parameters, so "one curve" must be an empirical claim.
- **Methods with substantial overlap:** base-seeded replay with a likelihood gate (TAR, 2610.24236); base-model archives for replay (EliteReplay, 2610.22290); Bayesian boundary gating (BBG, 2606.15455); off-policy assistance for zero-success groups (LSPO, 2607.27787; LUFFY); rollout allocation by posterior hit utility with per-problem normalization (HORA, 2605.07114); Knapsack RL (2509.25849); DARS (2508.13755), which adds rollouts for low-accuracy problems and plots Pass@1–Pass@K curves; MaxRL (2602.02710, ICML 2026). BBG gates on $\mathbb{E}[16(1-p)^{15}]$, whereas CM samples replay problems using $\mathbb{E}[16p(1-p)^{15}]$.
- **Not identified in this search:** a mean-preserving noise intervention; matched-pass@1 evaluation of advantage rules with intervals and registered decision rules on two families; the temperature control. DARS makes an informal matched-pass@1 comparison, without intervals or a registered protocol.

The literature review narrowed the novelty claim and motivated comparisons with the closest published methods.

21 Seeding the memory from the base model (Pre-registration 14)

Why we did it

Pre-registration 12 suggested a discovery limitation but did not establish it. Thirty-two independent samples find at least one success with probability $1-(1-p)^{32}$, about 87% at $p = 1/16$ and 64% at $p = 1/32$. Seeding the archive tests whether additional base-model discoveries improve replay.

Discovery. The base model sampled 32 answers for each of the 3,920 training problems (125,440 completions): 3,433 problems had at least one success, and 3,235 had a retained success before the trainer's exclusions. Retained answers must verify, terminate with a stop token and contain a parseable final box. After retokenization without special tokens, the answer must contain at most 511 tokens; the appended stop token completes the 512-token budget.

Arms (seeds 8–10, Pre-registration 12's settings): D32, seeded coverage-weighted memory at $\lambda = 0.5$ (candidate); U32, seeded uniform memory; S32, a supervised warm start on the same successes (AdamW 10^{-5}, 2.46 million response tokens) followed by Dr. GRPO; C05, unseeded coverage memory at $\lambda = 0.5$; G32, Dr. GRPO with groups of 32 (amendment 2); T32, TAR's gated replay on the same archive (amendment 4). A joint win requires a pooled gain $\geq +2$ points, a pooled 95% interval above 0, positive gains in every seed, and the 90% interval for step-150 pass@1 minus B contained in $[-1, +1]$. Claims must also survive Holm's correction over seven tests.

What we found

- Interim D32 against B's envelope: +1.02, 95% [+0.23, +1.68], per seed +0.71, +1.23, +1.12: average agreement. Step-150 pass@1 is 5.35 points below B, 90% [−5.99, −4.74]. No joint win.
- D32 against U32's trajectory: −0.39; against C05's: +0.92. Neither reaches the registered +1 threshold, so superiority of coverage weighting or seeding is not shown. Against S32's trajectory: insufficient overlap. Reported against B: U32 +1.37; C05 +0.33.
- At step 150, S32's three-seed mean pass@1 is 52.5 and pass@16 is 78.7. G32's mean pass@1 over available seeds is also 52.5; B's is 44.1. S32's pass@16 increase is concentrated on problems with no success in 32 base samples, from about 24 to 34 (Figure 10).
- T32's evaluation is pending.

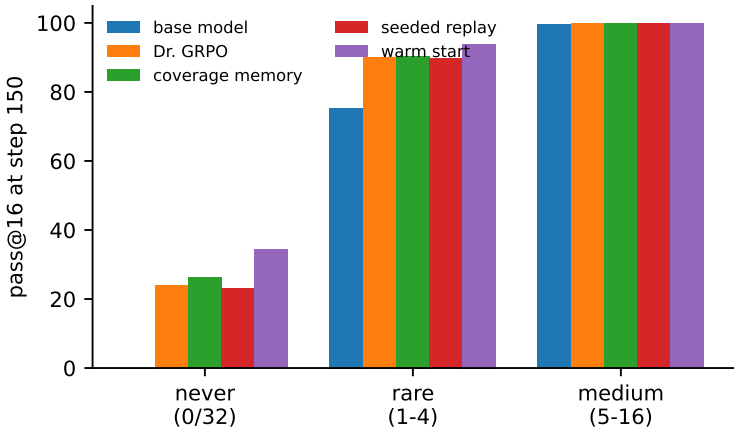


Figure 10: pass@16 at step 150 by the base model's successes in 32 samples on each test problem.

22 Checking the answer budget (Check 16)

Why we did it

At 512 tokens, 48% of Dr. GRPO's sampled answers at step 150 are truncated (S32: 33%; D32: 59%). If hard problems need longer answers, part of S32's gain could be "finishes within the budget".

What we found

Re-evaluated at 1,024 tokens (partial): the base model reaches pass@1 21.6 and pass@16 79.3 (36.2% truncated; 14.1 and 62.6 at 512); Dr. GRPO seed 10 at step 150 reaches 57.7 and 88.4 (15.5% truncated; 44.6 and 75.2 at 512); S32 seed 9 reaches 63.8 and 89.7 (4.9%). Absolute scores depend strongly on the response budget. The cross-seed pass@1 contrast is about 8.7 points at 512 tokens and 6.1 at 1,024; this descriptive comparison does not establish a budget mechanism. The registered check depends on S32's step-0 gain at matched pass@1 at both budgets and its 90% interval, which remain unreported. Frontier contrasts are computed separately within each budget.

23 2026 methods at matched accuracy (Pre-registration 15)

Why we did it

If the diagnosis is right, recent methods that report pass@k gains at fixed steps should also lie on the curve at matched pass@1. The audit also supplies the baselines a method must beat.

Arms (seeds 8–10, our recipe): BBG (W-REINFORCE with positive weight 0.3, gated by the posterior mean of $16(1-p)^{15}$, coefficients rescaled to mean 1 per step), W-REINFORCE (BBG's base, positive weight 0.3), MaxRL (advantages $(r-\bar{r})/\bar{r}$, none for zero-success groups); TAR is part of Pre-registration 14. Holm over the three directional tests.

What we found

Interim (training complete for most seeds, evaluation partial): BBG progresses much more slowly than Dr. GRPO (seed 9: pass@1 18.4, 24.8, 29.8 at steps 50, 100, 150). Its apparent gains (+1.3 to +2.6) lie between the base model and Dr. GRPO's step 50, where the registered reference is a straight chord that understates a concave trajectory. Check 20 adds Dr. GRPO checkpoints at steps 10–40 to measure that bias.

24 Pushing signal onto rare problems inside RL (Pre-registration 16, pilot)

Why we did it

The PI asked for a method inside RL rather than a supervised stage. Replay delivered far less push than the warm start, so this pilot tests stronger mechanisms on rarely solved problems.

Candidates (seed 12, 255 validation problems): AR, annealed replay (seeded coverage replay at $\lambda = 4$ falling to 0 by step 100, 16 problems replayed per step); GC, group completion (a stored success appended to a zero-success group as a ninth answer); PG, prefix-guided rollouts (for problems with at most 4 base successes in 32 and a stored success, two of eight rollouts continue from its first fraction $f_t = 0.5(1 - t/100)_+$); MXGC, MaxRL with group completion. References: B, S32, SD (S32 plus seeded coverage replay at $\lambda = 0.5$), MXR. Eligibility requires every registered

evaluation and at least two candidate checkpoints inside B's pass@1 range. Selection takes the largest matched-pass@1 gain $\geq +1.5$ among eligible candidates whose step-150 pass@1 is no more than 2 points below B. Candidates within 0.5 points of the largest gain are tied, with preference GC, PG, AR, then MXGC. This pilot ranks candidates; it does not establish an effect.

What we found

Interim: AR +1.30 and MXGC +1.12 at matched pass@1, but with step-150 pass@1 6.6 and 6.5 points below B, so neither qualifies. S32 is 8.5 points above B in pass@1 (beyond B's range). GC and PG are being evaluated; SD and MXR are training.

25 Does training transfer to rarely solved problems? (Check 17)

Why we did it

Under Dr. GRPO, a problem's own sampled gradient contribution is zero when all its rollouts fail. Updates on other problems can still change its success probability through shared parameters. This probe measures changes on held-out problems at different base success rates.

Probe. A fixed collection of 2,913 base-model traces covers 395 held-out problems: 149 with 1–4 successes and 246 with 5–16 successes in 32 samples at a 1,024-token budget. Each problem contributes up to four correct and four incorrect answers with parseable final boxes. At each checkpoint, $s_\theta(y \mid x) = |y|^{-1} \log \pi_\theta(y \mid x)$ is measured in nats per response token. The per-problem margin is $m_x = \bar{s}_{\text{correct}} - \bar{s}_{\text{incorrect}}$. Its change from the base model is a likelihood-margin proxy for transfer.

What we found

Four of six arm–seed pairs are available. At step 150, Dr. GRPO's rare/medium margin changes are +0.025/+0.025 (seed 8) and +0.023/+0.024 (seed 10). S32's are +0.049/+0.063 (seed 9) and +0.058/ + 0.070 (seed 10), with most of the increase already present after supervised training. These point estimates do not support a smaller margin change on rare problems. The registered pooled 90% intervals and length-matched analysis remain unreported.

26 A model of the frontier (Pre-registration 17)

Why we did it

The observed trajectories motivate a phenomenological model of per-problem success. We test whether base success counts, a frozen coupling parameter and each checkpoint's observed pass@1 suffice to predict pass@k. A successful fit supports this conditional description in the tested regimes; it does not establish that reward rules affect only training speed.

Model

$\theta_x(0) = \log \frac{c_x+a}{n-c_x+b}$ from the base model's counts (Beta(a, b) by moments), $\dot{\theta}_x = p_x(1-p_x) + \tau\overline{p(1-p)}$. A checkpoint is placed where the model's mean success equals its pass@1; the model then predicts pass@k for $k > 1$, strata and per-problem counts. $\tau = -0.20$ fitted once on Dr. GRPO's nine checkpoints, then frozen (code committed before any test).

Test set	checkpoints	pass@16 RMSE: model / uniform shift / affine	criterion (≤ 2)
Nine other rules, Qwen	56	**1.21** / 20.07 / 2.45	met
OLMo, five rules, τ from Qwen	50	**1.90** / 9.85 / 12.58	met
1,024 tokens	4	**0.85** / 11.48 / 1.27	met
Warm start, from its own step-0 counts	9	3.56 / 9.04 / 7.99	not met

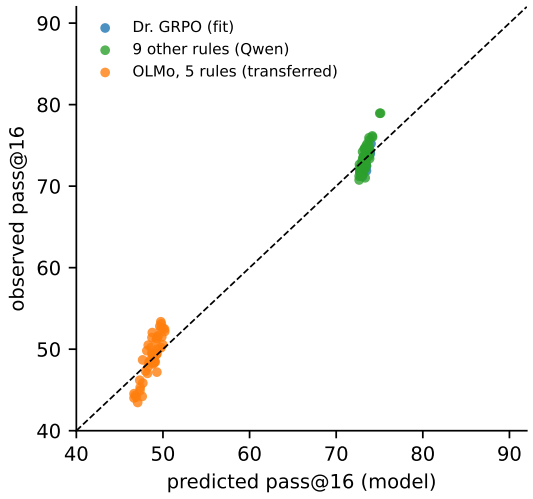


Figure 11: Predicted against observed pass@16 for the frozen model.

What we found

- Conditional on observed pass@1, aggregate pass@16 RMSE meets the registered $\leq$ 2-point criterion on T1, T4 and T5. The model also beats the affine alternative in per-problem log-likelihood on T1 and T4. Aggregate agreement masks OLMo stratum errors of +6.0 points for problems with no observed base success and −6.8 for rarely solved problems, measured as observed minus predicted pass@16.
- The warm-start test fails the registered RMSE criterion: 3.56 points. Its $c = 0$ stratum is 11.9 points below the prediction; refitted τ ranges from −0.65 to −0.70. These results show that the frozen model does not transfer to this regime.
- Partial deep sampling (three of four shards; 256 additional samples on 232 of 309 problems with $c \leq 4$) solves 53% of the previously never-solved problems at least once; their mean success rate is 0.93%. The refitted coefficient is $\tau_{\text{deep}} = +0.20$, 90% $[+0.10, +0.35]$, from 200 problem-bootstrap draws. This is a positive fitted coupling term, not a direct measurement of gradient transfer.
- The registered longer-training test has no reported verdict in `RESULTS_17_interim.md`. The 450-step reference has checkpoint results through step 250; an extrapolation verdict awaits an archived model-error analysis.

27 Larger groups: an interim comparison with longer training

Why we did it

Late S32 and G32 checkpoints exceed Dr. GRPO's 150-step pass@1 range. A reference run planned for 450 steps extends the range for descriptive comparisons.

What we found

Dr. GRPO seed 13 reaches (pass@1, pass@16) = (36.6, 72.0), (40.3, 71.1), (43.5, 74.1), (48.3, 76.5), (51.7, 78.6) at steps 50–250. Mean matched-pass@1 differences against this trajectory are: B seeds 8–10 +0.47, G32 +0.60, S32 +2.38, CM +1.73, U32 +1.74, D32 +1.06, C05 +0.82. Only 4 of 6 G32 checkpoints and 10 of 12 S32 checkpoints are in range; the four replay arms have 9 of 9 each. These exploratory contrasts use one reference seed and have no reported confidence intervals. At step 250, B uses 32,000 rollouts and reaches pass@1 51.7; at step 150, G32 uses 76,800 and reaches 52.5. The endpoints do not match accuracy exactly. Larger groups progress faster per optimizer step, but a frontier improvement or equivalence is not established. Steps 300–450 and reference seeds 14–15 are pending.

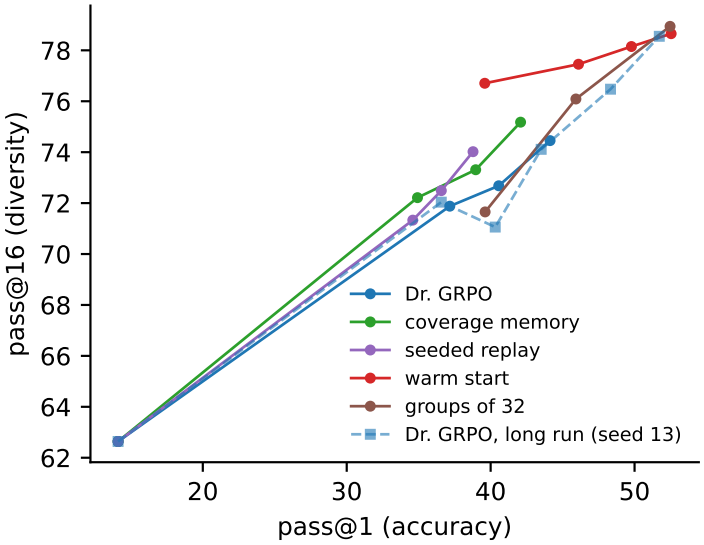


Figure 12: Seed-mean trajectories on Qwen, with the long Dr. GRPO run (dashed).

28 Allocating rollouts by frontier value (Pre-registration 18)

Why we did it

A problem's own Dr. GRPO gradient contribution requires a mixed-reward group (Lemma 1). Allocating more rollouts to rarely solved problems raises the chance of observing a success there; the frozen frontier model forecast that allocation by frontier value (AC) would move the curve by about +4 points.

Lemma 1 (stated for reference)**.** *For n independent rollouts with success probability p, the sampled Dr. GRPO gradient contribution of a problem is zero when all rewards are identical. A mixed-reward group occurs with probability $1-p^n-(1-p)^n$, approximately np when $np \ll 1$. Finite multiplicative reweighting of the centered advantages preserves this zero-gradient condition. Updates through shared parameters can still change the problem's success probability.*

Design (seeds 8–10; 16 problems and exactly 128 rollouts per step): 4 pilot rollouts per problem, 64 extras, at most 32 per problem. AC allocates extras by frontier value, AV by $\mathbb{E}[p(1-p)]$, HR adapts HORA's hit utility.

What we found

- AC against B's envelope: +0.14, 95% $[-0.36, +0.58]$; average agreement (rule 1 needed $\geq +2$). AC against AV: +0.51 $[+0.04, +0.99]$; AC against HR: +0.00 $[-0.52, +0.46]$; neither is Holm-significant.
- HR and AV forecasts met their registered criteria; AV's frozen-forecast error was 0.90 points. AC's observed mean was −0.05 against +4.12 forecast. Success-bearing pilot-zero groups were 50–54% under AC against 18–21% under AV, yet AC stays on the curve.
- The starvation explanation (the curve is fixed because rare problems get no signal) is not supported in this form.

29 External versus self-generated information (Pre-registration 19) and the answer budget (Checks 16 and 21)

What we found

- P19 rule 1 (human solutions on the 487 never-solved problems, HQ against HP) is not interpretable: after fine-tuning, RL moves pass@1 by at most 1.6 points, so the trajectories do not overlap. On never-solved problems, HQ minus HP is +0.58 $[-2.13, +3.43]$; 434–435 of 487 solutions survived the filters.
- At 512 tokens, the human-solution arms lay 5–6 points above B's curve, but only 6–7% of their answers were truncated against 48% for B (an answer-budget confound). Check 16 (the base-success warm start S32): its step-0 gain falls from +4.77 $[+3.45, +6.07]$ at 512 tokens to +2.15 $[+1.10, +3.20]$ at 1,024 (45%): *largely a budget effect*; a smaller gain (interval above 0) remains.
- Check 21 (partial, interim): the human-solution warm start HP at 1,024 tokens, seed 10, lies 2.4–3.9 points *below* B's curve; the other seeds were not evaluated (their jobs produced no checkpoints).

30 Early reference checkpoints (Check 20)

Registered as a sensitivity analysis of B's early trajectory (steps 10–50). No result is recorded; the registered verdicts do not depend on it.

31 Moving to AWS, and a faster evaluator

All compute moved to one AWS account (SageMaker): L40S machines for single-GPU work and 8×A100 machines (ml.p4de.24xlarge, $31.56 per hour) packed to use every GPU. The Hugging Face evaluator needed about 14 hours per 7B checkpoint; four such evaluations were stopped after about 27.5 A100-hours (about $870) without output. A vLLM evaluator (`eval_vllm.py`) replaced it: identical prompts, stopping and scoring, per-problem sampling seeds, a tokenizer compatibility layer, an isolated environment, and about 13 minutes per 1.5B and 30 minutes per 7B checkpoint.

Its agreement with the earlier evaluator was checked before use (Pre-registration 20, amendment).

32 A larger model: Qwen2.5-Math-7B at 2,048 tokens (Pre-registration 20)

Why we did it

A diagnosis at one 1.5B model could be a small-model artifact. The core contrasts were repeated at 7B with a longer answer budget, with a frozen forecast from the 1.5B frontier model.

Design. Qwen2.5-Math-7B, multi-GPU trainer (FSDP2, 4 A100 per run, fp32 weights with bf16 autocast; parity with the single-GPU trainer checked at 1.5B), seeds 8 and 9, 150 steps, 16×8 rollouts, 2,048 tokens; arms B7, PSR7, NSR7, AC7; 511 test problems and a harder 253-problem cohort.

What we found

- PSR7 against B7: -0.09, 90% $[-0.48, +0.31]$; NSR7 against B7: -0.09 $[-0.45, +0.27]$: both average equivalence. AC7 against B7's envelope: -0.15 $[-0.57, +0.16]$: the allocation null replicates. NSR7 against PSR7 ($+0.16$) is not interpreted (13.7% of bootstrap points clamped in seed 9).
- The frozen frontier model predicts B7's pass@16 with RMSE 0.23 points over 8 seed-steps (uniform shift 5.53, affine 0.93): supported. Its stratum errors are large on never-solved problems ($+7$ to $+23$) and on $c = 1$–4 (-3 to -7).
- On the harder cohort the point estimates are within ± 0.9; two contrasts are uninterpretable and one is inconclusive. Greedy accuracy 74–76% for all arms.

33 A larger answer budget: 1.5B at 1,024 tokens (Pre-registration 21)

What we found

- NSR1k against PSR1k -0.25 $[-0.72, +0.20]$; PSR1k against B1k $+0.11$ $[-0.26, +0.49]$; NSR1k against B1k -0.15 $[-0.62, +0.31]$; AC1k against B1k's envelope -0.12 $[-0.60, +0.29]$ (95% intervals). All four reach average equivalence: **the core result replicates at 1,024 tokens**.
- Step-150 truncation 15.2–17.0% in every arm (below the 20% qualifier); greedy accuracy 63.6–64.6%.

34 Guidance with human solutions inside RL (Pre-registration 22)

Why we did it

Pre-registration 19's human solutions acted through a supervised stage. Here they guide RL directly: some rollouts continue from a prefix of a human (HG, HGm) or a stored self-generated (PG) solution on rarely solved problems.

What we found

- Rule 1, HG1k against B1k's envelope: -0.05, 95% $[-0.52, +0.37]$: on the curve (average equivalence).
- Rule 2, HGm1k (human prefixes on the problems where a self-generated success also exists) against PG1k: $+0.05$; not interpreted (15.8% clamped in seed 9).
- Reported: PG1k against B1k -0.11 $[-0.55, +0.32]$; HGm1k against B1k -0.01 $[-0.41, +0.37]$.

35 Minimum-help credit (Pre-registration 23, pilot; 24 and 25)

Why we did it

If rare problems fail because a long solution must be found from scratch, crediting the policy's own failed prefixes by the smallest human hint that makes them succeed gives dense, problem-specific signal.

What we found

MH against B1k's envelope on 255 validation problems: -0.01, 95% $[-1.24, +0.89]$; the averaged-value control AVH -0.05; MH against AVH $+0.05$. The credit operated at the registered dose (520 credited prefixes; contrast in 61% of probed problem-steps). Neither arm reached the $+1.5$ selection gate: **dropped**. The conditional 7B test (Pre-registration 24) was withdrawn; the causal update-direction design (Pre-registration 25) was drafted but not run.

36 Do methods move along one shared direction? (Pre-registration 26)

Why we did it

Different methods give the same curve. Reported rank-1 structure of RLVR weight updates (AlphaRL, RELEX) suggested a hypothesis: every method moves the weights along one shared direction, so methods can only change speed. Step 1 measured this; step 2 piloted four methods that change the update's direction or add a new gradient source.

Weights (step-150 checkpoints of 11 runs, 196 weight matrices; saved in bf16):

- Rank-1 approximations capture 4.7–5.6% of the update energy; the top singular directions align across runs (0.82–0.91 with B1k8).
- Update cosines: Dr. GRPO seed to seed 0.34; methods against B1k8 0.34–0.47; PSR1k8 against NSR1k8 0.27 (the least similar pair). The registered support threshold (0.5) was not met.
- Split checkpoints of B1k8's update: base 21.6/78.0 (pass@1/pass@16); trained non-matrix parameters only 23.4/79.1; rank-1 part 31.8/84.8 (distorted by bf16 rounding: cosine 0.35 with the intended part); remainder 57.5/86.9; full B1k8 58.2/86.5. The remainder retains about 98% of B1k8's pass@1 gain over "other"; the bf16-distorted rank-1 checkpoint retains 24%.

Gradients (32 problems, two independent groups of 8 answers each, CountSketch 2^{20}): the same problem's gradients from the two groups have mean cosine 0.01; gradients of different problems have mean cosine about 0; the cross-group Gram matrix has 37% negative spectral mass, so the registered whitening reading is inconclusive.

Pilots (seed 8, 1,024 tokens, 255 validation problems; selection needs $\geq +1.5$ at matched pass@1, pass@1 within 2 points of B1k, and an active mechanism):

Arm	Change	Gain [95%]	Mechanism
OD	remove 75% of each update's component along the tracked top direction, renormalize	+0.20 $[-0.84, +1.19]$	share 0.051; realized cosine 0.986 (gate 0.98 not met)
PW	whiten problem weights with the sketched gradient Gram matrix	−0.04 $[-1.21, +1.05]$	active: alpha ratio > 1.5 in all steps, cosine 0.948
TD	add the log-likelihood of verified 7B solutions on base-rare problems	+0.31 $[-1.05, +1.35]$	6.55 teacher solutions per step (1,580 of 1,668 rare problems covered)
RTD	TD with the teacher gradient projected off the tracked direction	+0.22 $[-0.97, +1.17]$; vs TD −0.03	15% of the teacher gradient removed

What we found

No pilot qualifies at the tested dose, including TD, which adds verified teacher solutions for hard training problems. The saved-update shared-direction criterion fails, while expected-gradient structure remains unresolved.

37 Same problems, same rates: where the gain comes from (Investigation 27, exploratory)

What we found

- **Per problem.** At matched pass@1 (within 1.5 points), two different methods differ per problem by about as much as two Dr. GRPO seeds (RMS 6.6–6.7 points). A paired excess statistic of squared per-problem distance, as a share of the seed-to-seed squared distance (42.0 pp^2): all methods +2.1% [−1.9%, +6.1%]; PSR +7.5% [+2.6%, +12.4%]; NSR +7.0%; hint and prefix arms about 0; PG −7.1%. Advantage-rule changes leave a small per-problem signature; arms that keep Dr. GRPO's advantages stay within seed variation.
- **By difficulty.** Success by base count is similar across methods at matched pass@1 (at pass@1 $\approx$ 58: 9% on never-solved problems, 33% at $c = 1$–2, 67% at $c = 3$–8). A uniform logit shift of the base rates fits poorly (pass@16 RMSE 10 points over 71 evaluations), while the frozen frontier law fits with 0.82 points over the 65 $T = 1$ checkpoints.
- **Finishing versus accuracy** (1.5B, 1,024 tokens). Base: 21.6% right, 35.8% boxed but wrong, 42.5% without a final box (35.8% truncated). B1k8 step 25: 40.0 / 35.8 / 24.2; step 150: 58.1 / 30.4 / 11.5. Exploratory accounting assigns +11.6 of the +35.8-point mean step-150 gain (mean over runs) to increased boxing at the base's boxed-answer accuracy; this is not a causal estimate. Accuracy among boxed answers rises from 37.6% to 65.7%. Format errors are 1–2%.
- **What stays unsolved.** Across problems, step-150 success is described by $0.52 + 0.98\, r_i + 0.24\, u_i - 0.70\, w_i$ (base right, unfinished and boxed-wrong rates; descriptive): the base's boxed-wrong mass is negatively associated with post-RL success.

38 Comparison with the budgeted-search law (restricted)

The closest prior work (2609.01274) maps an RL model's pass@k to the base model's pass@N with $N = \alpha k^{\beta}$. Fitted on Dr. GRPO's evaluations ($\beta = 0.845$) and restricted to one decoder with 32 base samples, it can score only 8 of 53 held-out evaluations (the base's pass@32 is below most trained checkpoints' pass@16). On those 8, Dr. GRPO's curve predicts held-out pass@16 with RMSE 0.63 points against 1.93 for the restricted law. A full comparison needs the base with 256 samples (Pre-registration 28).

39 Running now

Pre-registration 27 (reason 3: is the training too short?). Two Dr. GRPO runs of 1,000 steps (about 4 passes over the training problems) at learning rates 10^{-6} (LH1) and 3×10^{-6} (LH3), 1.5B, 1,024 tokens. Primary: a step-1,000 pass@256 gain of at least 1 point over the base (both 256 samples), with a Holm-adjusted 95% interval above zero. Reported: pass@16 against the frozen law, and the change in the base's difficulty order. Started October 28; both runs passed step 800.

Pre-registration 28 (evaluation only). 21 published RLVR checkpoints trained from Qwen2.5-

Math-1.5B or 7B (Oat-Zero, SimpleRL-Zoo and -Zero, LUFFY, ExGRPO, ReLIFT, RELEX, one-shot RLVR, RLHFlow GRPO/DAPO/RAFT++, Zero-Solver, correct- and random-reward reproductions) evaluated under one protocol and placed against Dr. GRPO's curve and the frozen law; a forced final answer for unfinished samples of the base and B1k8; the base with 256 samples and at lower temperatures.

40 Where we stand

40.1 What is established

1. Compared at matched pass@1, most interpretable comparisons show average agreement with Dr. GRPO's pass@16-versus-pass@1 curve within the tested regimes and observed support; overlap limitations, failed pilot selections and S32's residual gain qualify this. The interventions are: advantage rules (PSR, NSR, constant baselines, pass@k objectives, MaxRL, BBG), entropy terms, replay and group completion, larger groups, rollout allocation (AC, AV, HR), supervised warm starts once the answer budget is controlled, human-solution prefixes, minimum-help credit, update-direction changes, problem whitening, and teacher distillation at the tested dose. The pattern holds at 1.5B with 512 and 1,024 tokens and at 7B with 2,048 tokens (three of four 7B contrasts interpretable).
2. At matched pass@1, methods show similar per-problem profiles (exploratory), although their saved weight updates differ; a frozen per-problem law from the base model's counts predicts the curve (0.23–0.82 points RMSE).
3. By exploratory accounting, about a third of the 1,024-token gain is associated with finishing answers; the base's boxed-wrong mass is negatively associated with post-RL success.

40.2 Explanations ruled out or weakened

Starvation of rare problems (allocation raised success-bearing pilot-zero groups from 18–21% to 50–54% without moving the curve); external information as the key (human solutions, prefixes, teacher solutions); a shared saved-weight direction (cosines 0.27–0.47; the remainder outside each matrix's top direction retains about 98% of the pass@1 gain).

40.3 Open

Is the lock a short-horizon effect (Pre-registration 27)? Do published checkpoints lie on the curve (Pre-registration 28)? How much of RL's gain is a decoding fix (forced answer)? Does distillation at a realistic scale move the curve?

40.4 To ICML (late January 2027)

November: Pre-registrations 27 and 28; distillation dose test; PBA, ReCo and BBG on the curve; paper draft v5. December: confirmation at 7B and on the fresh Omni-MATH set; full budgeted-search comparison. January: writing and internal review.

A Methods reference

Rule	Definition	Settings
Dr. GRPO (B)	$A_i = r_i - \bar{r}$ within a group of n; loss $-(PnL)^{-1} \sum_i A_i \sum_t \log \pi(y_{it})$	$P = 16$, $n = 8$, $L = 512$, AdamW 10^{-6}, 150 steps, on-policy
PSR / NSR	$A = r$ / $A = r - 1$	as B
W-REINFORCE	$A = w$ if correct, -1 if incorrect	$w = 0.1$ (Zhu et al.) or 0.3 (BBG)
Constant baselines	$A = r - b$	$b \in \{0, 0.25, 0.5, 0.75, 1\}$
pass@k objective	unbiased pass@4 weights with leave-one-out baseline	$k = 4$, $n = 8$
PSR + noise	$A_i = r_i + c_t s_i$, $s_i = \pm 1$	c_t^2 matches NSR's predicted second moment
Unlikeliness reward	correct samples in mixed groups get $1 - \beta(G - \text{rank})/G$	$\beta = 0.25$
Coverage memory (CM)	add $\lambda \nabla \sum_q \log \sum_{y \in S_q} \pi(y \mid q)$ over 8 archived problems sampled $\propto \mathbb{E}[k\, p(1-p)^{k-1}]$, $p \sim \text{Beta}(c + \frac{1}{2}, n - c + \frac{1}{2})$	$\lambda = 1$, $k = 16$, ≤ 4 successes per problem
Uniform memory (M)	as CM with uniform sampling	$\lambda = 1$
Seeded memory (D32, U32)	archive initialized with one base success per problem; retired at the first own success	$\lambda = 0.5$
Warm start (S32)	mean token NLL on base successes, then B	AdamW 10^{-5}, batches of 8, 2.46M tokens, clip 1, optimizer reset
Groups of 32 (G32)	B with $n = 32$	512 rollouts per step
T32 (TAR adaptation)	$\lambda \,\text{mean}_y\, \text{sg}[g] \sum_t \text{sg}[w_t](-\log \pi_t)/(\epsilon + \sum_t \text{sg}[w_t])$; $\phi(z) = \max(1 - e^{-z}, 0)$, $g = \phi(\bar{\delta})$, $w_t = \phi(\delta_t)$, $\delta_t = \log \pi_{\text{base},t} - \log \pi_t$; sg stops gradients and $\bar{\delta}$ averages response tokens	$\lambda = 1$, $\epsilon = 10^{-8}$; 16 uniformly sampled base traces per step, no retirement; shared discovery bank
BBG	W-REINFORCE times $\alpha(m) = u(m; 8, 16)\mathbf{1}[u \geq 0.09]$, $u = 16 \prod_{j=1}^{15} \frac{8-m+j}{9+j}$, rescaled to mean 1	$w = 0.3$
MaxRL	$A = (r - \bar{r})/\bar{r}$; 0 if $\bar{r} = 0$	as B
Annealed replay (AR)	seeded coverage replay, $\lambda_t = 4(1 - t/100)_+$	16 problems per step
Group completion (GC)	append a stored base or policy success to a zero-success group; center rewards over the resulting $n + 1$ answers	off-policy likelihood contribution; no importance weight
MXGC / SD / C05	MXGC: MaxRL with group completion; SD: S32 followed by seeded coverage replay; C05: online CM without base seeding	SD and C05: $\lambda = 0.5$
Prefix-guided (PG)	2 of 8 rollouts continue from fraction f_t of a stored success; prefix tokens are context with no gradient; verify prefix plus continuation	$f_t = 0.5(1 - t/100)_+$; $c \leq 4$ of 32; prefix plus continuation ≤ 512 tokens

Rule	Definition	Settings
Allocation (AC, AV, HR)	pool 4 pilot rollouts per problem with 64 extras; AC/AV use posterior $\text{Beta}(c+\frac{1}{2}, 4-c+\frac{1}{2})$, HR uses $\text{Beta}(c+1, 5-c)$, where c counts pilot successes	128 rollouts total, ≤ 32 per problem; fixed loss denominator $16 \cdot 8 \cdot 512$; HR additionally scales each problem's terms by $8/N_i$; AC uses largest-remainder rounding and uniform fallback when no positive-value capacity remains

Training and archived data. The online RL component uses one update on fresh rollouts, without PPO clipping or a KL penalty; replay and group completion add off-policy terms. D32 priorities start from discovery counts $(c, 32)$ and subsequently use the latest online group; U32 samples uniformly. Base traces retire when the first retained policy success is stored. S32's 2.46-million-response-token target may overshoot by one batch of 8 answers. Human-solution training withholds 16 calibration problems per seed and applies length and re-verification filters; retained counts, actual token totals and epochs must be reported. T32 and BBG are recipe adaptations: T32 uses the shared discovery bank, and BBG rescales coefficients over 16 problems without accumulating the paper's 512 active prompts.

Evaluation. Main Qwen comparisons use 511 audited held-out MATH problems, 32 samples at $T = 1$, and unbiased pass@k; the pilot uses 255 validation problems. Greedy accuracy uses 3,382 problems. The primary frontier metric is pass@16. Reference curves use linear interpolation after sorting by pass@1, with ties resolved by step or temperature. B's envelope takes the larger defined value of its training trajectory and its step-150 temperature trajectory ($T = 1, 1.15, 1.3$). Gains average qualifying checkpoints within each seed, then average seeds; at least two checkpoints per seed must overlap the reference. Reference curves are recomputed in each of 2,000 paired problem-bootstrap draws, and endpoint-clamping frequencies are reported. Draws are comparison-specific in Pre-registration 12 and shared where specified in later registrations. Intervals condition on the training seeds and shared discovery archive. The reused test set's second half is not untouched confirmation; registered confirmation uses 512 fresh problems when triggered.

Frontier model. The model in Section 26 fits $\tau_{32} = -0.20$ and affine slope $\beta = 2.10$ using only B's nine fitting checkpoints. Uniform-shift and affine comparators are $\theta'_x = \theta_x + \alpha$ and $\theta'_x = \alpha + \beta\theta_x$, with α matching observed pass@1. The model's evolution parameter also matches observed pass@1. Registered RMSE categories are supported at ≤ 2 points, partial at 2–3, and falsified above 3; T1–T4 also require higher total per-problem log-likelihood than the affine comparator. The reported τ_{deep} interval uses 200 problem-bootstrap draws with refitting.

B Experiment tracker (Pre-registrations 12–28)

#	Question	Design	Result
12	Does coverage memory move the curve?	4 arms, 3 seeds, 511 problems	+1.19 [+0.54, +1.79]: average agreement

#	Question	Design	Result
13	OLMo replication of the method	planned	not run (the method did not pass)
14	Base-seeded memory and controls	D32, U32, S32, C05, G32, T32; 3 seeds	D32 +1.02, slower; S32 and G32 beyond B's range; T32 pending
15	2026 methods at matched pass@1	BBG, W-REINFORCE, MaxRL; 3 seeds	interim: BBG slow, apparent gains in the chord region
16	Push mechanisms inside RL (pilot)	AR, GC, PG, MXGC + refs; 1 seed	interim: AR, MXGC not eligible; GC, PG pending
C16	512 vs 1,024 tokens	re-evaluation	partial: all values rise at 1,024
C17	Transfer to rare problems	likelihood margins of fixed base answers	4/6 arm–seed pairs: similar rare/medium point estimates for B; pooled intervals unreported
17	Frontier model	frozen model, 5 test sets	aggregate RMSE criterion met on 3; warm-start test fails; longer-training analysis unreported
C20	Early reference checkpoints	B steps 10–50, 3 seeds	no result recorded
18 (final)	Allocation by frontier value	AC, AV, HR; 3 seeds	AC +0.14: on the curve; AC forecast not confirmed
19 (final)	External vs self-generated information	HQ, HP, HS	rule 1 not interpretable (no overlap); 512-token gains are budget effects (Check 16)
C21	Human-solution warm start at 1,024 tokens	HP seed 10	partial (interim): 2.4–3.9 below the curve
20	7B at 2,048 tokens	B7, PSR7, NSR7, AC7; 2 seeds	PSR7 and NSR7 equivalent to B7; AC7 null; frozen law RMSE 0.23
21	1.5B at 1,024 tokens	B1k, PSR1k, NSR1k, AC1k; 3 seeds	all four contrasts equivalent: replicates
22	Human/self prefixes inside RL	HG1k, HGm1k, PG1k; 3 seeds	HG1k on the curve; rule 2 not interpreted
23	Minimum-help credit (pilot)	MH, AVH; seed 8	no candidate; dropped
24, 25	7B MH test; causal direction test	drafts	withdrawn; not run
26	Shared direction; OD, PW, TD, RTD pilots	diagnostics + 4 pilots, seed 8	shared-direction criterion not met; no pilot qualifies
I27	Per-problem agreement; finishing	existing evaluations	methods match seeds per problem; a third of the gain is finishing
27	Long and stronger training	LH1, LH3; 1,000 steps	running
28	Published checkpoints; forced answer	21 checkpoints; evaluation only	launching

C Compute and engineering

Since October 25 (version 5). All runs use one AWS account: SageMaker L40S machines (single GPU) and 8×A100 machines packed so that every GPU has work (several runs and evaluations per machine; lanes are described in `experiments/sagemaker/plans/`). Registered experiments are reviewed by Codex before launch, and unregistered exploratory analyses are labelled as such; code, plans, job lists and results are in the repository. Known one-off losses: 27.5 instance-hours (220 A100 GPU-hours, about $870) on Hugging Face 7B evaluations that were too slow (replaced by vLLM), duplicate 7B evaluations in two regions (about 7 hours of one machine), and a teacher-generation job that failed after 16 minutes (about $8).

Earlier. The registered pilot uses Tamia H100s for every arm and reference. Check 20 uses Nibi H100s and compares step 50 with the earlier L40S reference. The longer-reference design places seed 13 on AWS and seeds 14–15 on Tamia. A reproducible compute comparison still needs per-run GPU-hours, software versions, optimizer and numerical settings, and attempted, completed and skipped optimizer-update counts. Accounting must include discovery, supervised training, rollout generation, replay and evaluation. All code, registrations, results and figures: `github.com/mayman150/rlvr-adaptive`.